

Advanced Topics in Computer Graphics I

Introduction



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Institute of Computer Science II - Computer Graphics
University of Bonn

April 13, 2026





Important Information:

- ▶ 9 credit points, 4h lectures, 2h exercises
- ▶ Lecture times:
 - ▶ Tuesday and Thursday 12:15 - 13:45
- ▶ Lecturer:
 - ▶ Prof. Dr. Reinhard Klein
- ▶ Tutors:

Domenic Zingsheim: `zingsheim@cs.uni-bonn.de`



- ▶ Distribution of exercise sheets and lecture slides via eCampus:
 - ▶ `https://ecampus.uni-bonn.de/goto.php?target=crs_4053074&client_id=ecampus`



- ▶ Weekly exercise sheet available on eCampus.
- ▶ Form groups of two to three people!

Theory

Practice

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- | | |
|--|---|
| <ul style="list-style-type: none">▶ Write solutions in $\text{\LaTeX}$▶ Hand in single PDF file on eCampus | <ul style="list-style-type: none">▶ C++ / CUDA / OptiX▶ Develop on your PC or in CG pool |
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- ▶ Questions?
 - ▶ Ask on eCampus
 - ▶ Tutors available via email
 - ▶ Admission to exam:
 - ▶ At least 50% in theoretical exercises **and**
 - ▶ At least 50% in practical exercises
 - ▶ For 70% of the exercise sheets, you have to acquire at least 50% of the total points



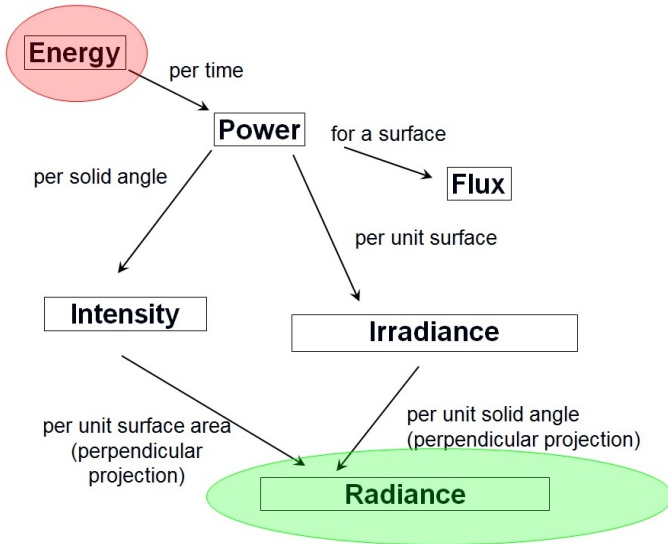
- Please vote on the poll on eCampus and fill in **all possible dates** where you are available.

	Mon	Tue	Wed	Thu	Fri
8:00 – 10:00					
10:00 – 12:00	Exercise?	Exercise?		Exercise?	
12:00 – 14:00		Lecture		Lecture	Exercise?
14:00 – 16:00					

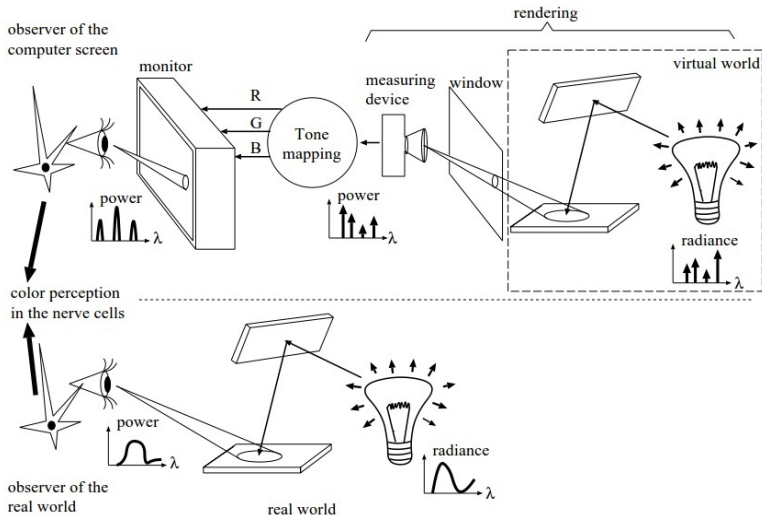


Content Overview



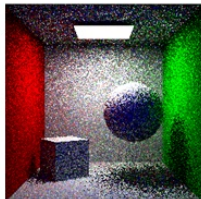


Chapter 2: Global Illumination Problem

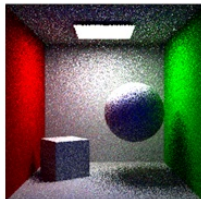




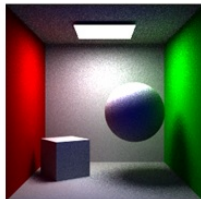
Chapter 3: Monte Carlo Integration



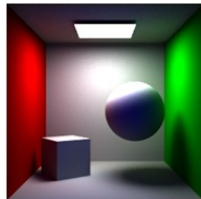
1 sample/pixel



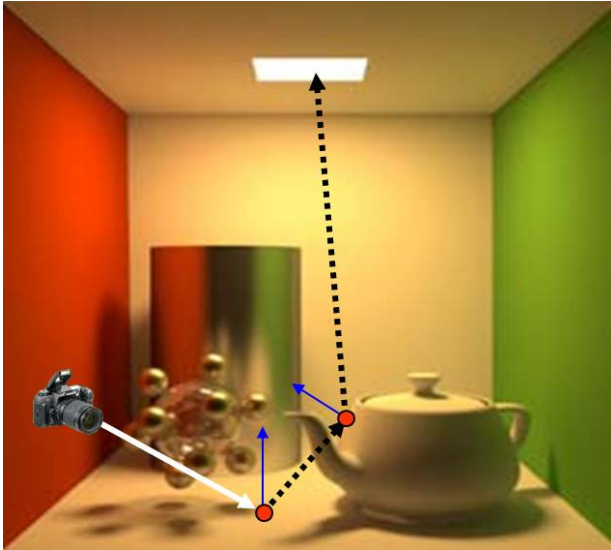
4 samples/pixel

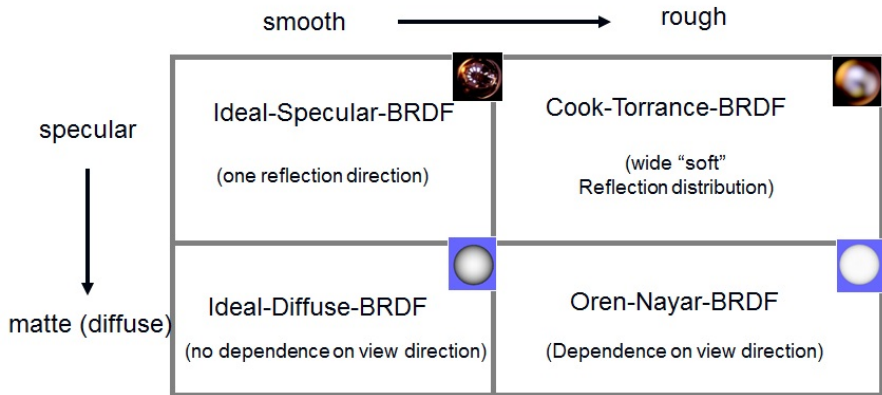


64 samples/pixel



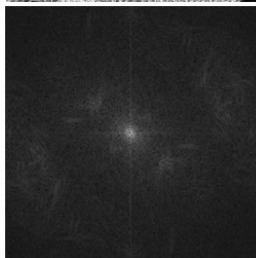
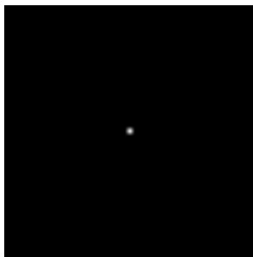
1024 samples/pixel



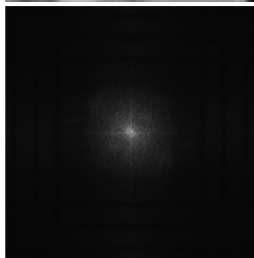
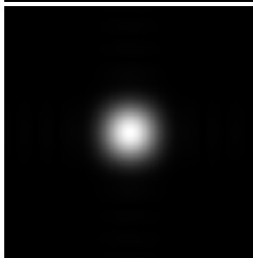


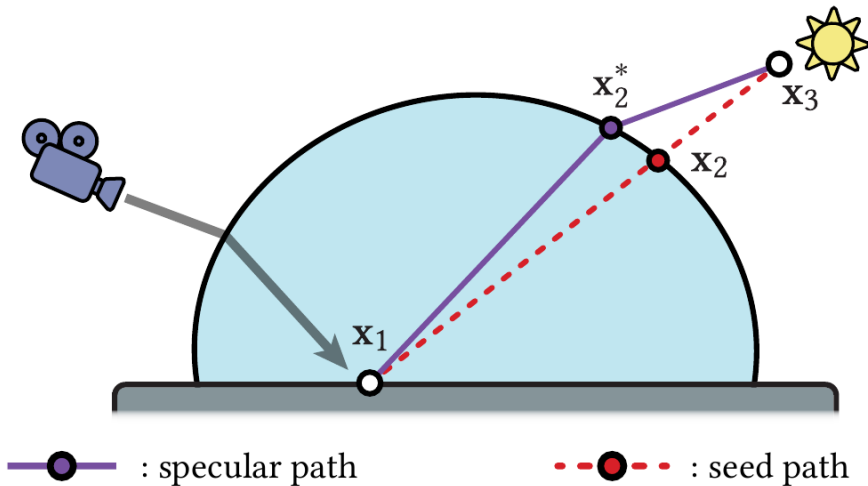


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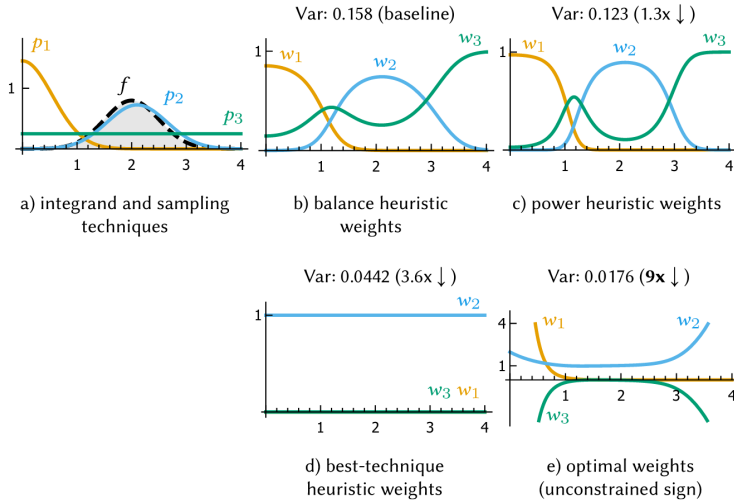
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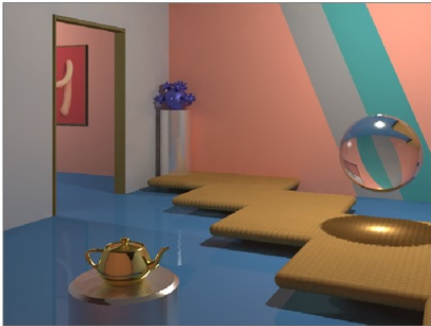


Chapter 8: Optimal MIS



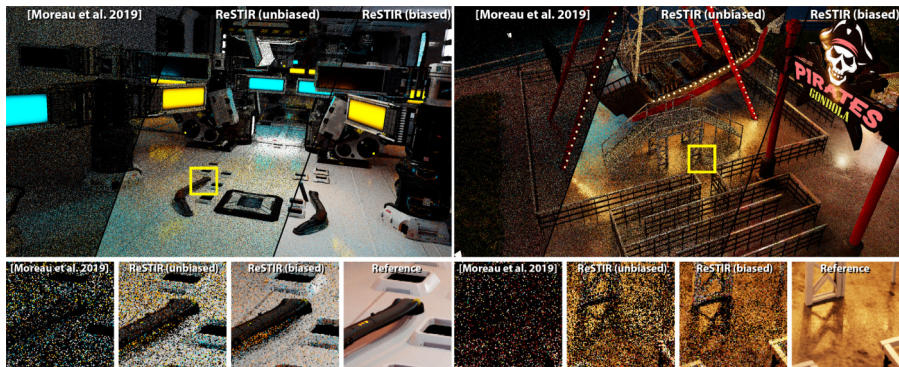


Chapter 9: Photon Mapping





Chapter 10: Generalized Resampled Importance Sampling

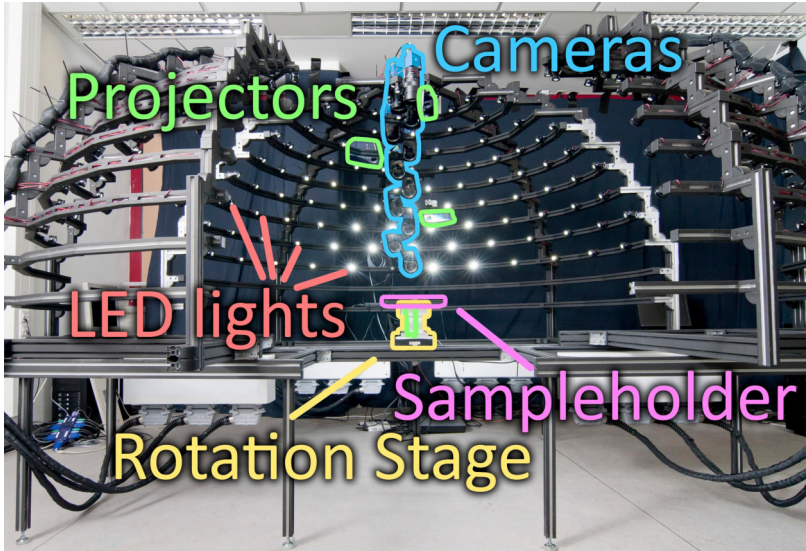


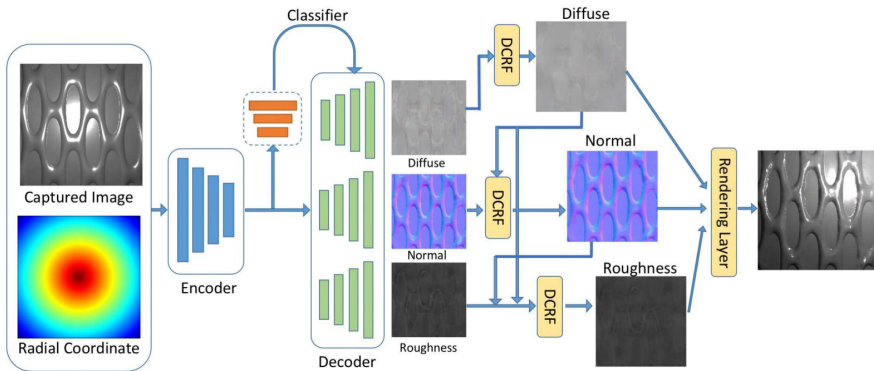


Rendering at Low Concentrations



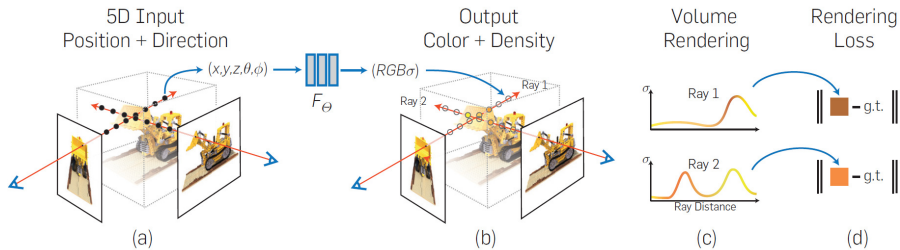
Rendering at Natural Concentrations







Chapter 14: Neural Rendering



Chapter 15: Differentiable Rendering

